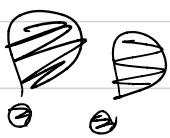
30.05

$G_1 = (\Sigma, N_1, S_1, P_1)$ $\quad S \notin N_1 \cup N_2$

$G_2 = (\Sigma, N_2, S_2, P_2)$



$N_1 = \{M, N\}$

$N_2 = \{P, Q, M\} \xrightarrow{\text{ака } M_1 \cap N_2 \neq \emptyset} N_2 = \{P, Q, M_1\}$

Съобсяване на правилата : дели. ≤ 2

$A \to a_1 a_2 \dots a_n \in (\Sigma \cup N)^*$

$\downarrow$

$A \to a_1 A_2, A_2 \to a_2 A_3, \dots, \underbrace{A_{n-1} \to a_{n-1} a_n}_{A_i \notin N}$

$A_1 \to a_1 A_2 \to a_1 a_2 A_3 \to \dots \to a_1 \dots a_{n-2} A_{n-1} \to a_1 \dots a_n$

① Нека L-рег., $\Sigma = \{a, b\}$

$\Rightarrow L_{sort} = \{a^{|w|_a} b^{|w|_b} \mid w \in L\}$ е КСЕ

$A = (\Sigma, Q, q_0, \Delta, F)$ - ТДКА : $L(A) = L$

$G = (\Sigma, Q, q_0, P)$, $P = \{f \to \varepsilon \mid f \in F\} \cup$
$\{q \to ap \mid \delta(q,a) = p\} \cup$
$\{q \to pb \mid \delta(q,b) = p\}$

$L(G) \stackrel{?}{=} L_{sort}$

($\subseteq$) Нека $\alpha \in L(G)$

$q_0 \to \alpha_1 \to \alpha_2 \to \dots \to \alpha_n$, където $\alpha_1 \dots \alpha_{n-1}$ съдържа нет. и $\alpha_n \in \Sigma^*$

Съобразете се $p \in P \Rightarrow$ за p може само да нарява или задържа броя нетерм, но да ги увеличава.

$\Rightarrow$ за $q_0 \to \alpha_1$ и $\alpha_i \to \alpha_{i+1}$, $i \in \{1, \dots, n-2\}$ използв. правила от вида $q \to ap$ и $q \to pb$ и за $\alpha_{n-1} \to \alpha_n$ $f \to \varepsilon$, $f \in F$.

Нека за $i_1 \dots i_k$ $\alpha_{i_j} \to \alpha_t$ използва правилото $q \to pb \Rightarrow \delta(q_0, a_{i_1}) = p$, $\delta(p, a_{i_1}) = p_1 \dots p_n$, където $a_{i_j} = b$, а $a_t \neq a_{i_j} = a$

$\delta(\delta \dots \delta(q_0, a_{i_1}), a_{i_2}) \dots a_{n-1}) = f \in F$

$\Rightarrow a_{i_1} \dots a_{n-1} \stackrel{\beta}{=} \in L$

$|\beta|_b = |\{i_1, \ldots, i_k\}|$ = броя прилагания на правилото $q \to pb$

$|\beta|_a = |\beta| - |\beta|_b$ = броя прилагания на $q \to ap$ и освен това:

$q_0 \to \ldots \to a^t p b^t \to \ldots \to a^n b^m$ и $|a^n b^m|_a = |\beta|_a$

$|a^n b^m|_b = |\beta|_b$

$\Rightarrow \alpha_n \in L_{sort}$

(⊇) Нека $\alpha \in L_{sort} \Rightarrow \exists \beta \in L: \alpha = a^n b^m$ и

$|\beta|_a = n$ и $|\beta|_b = m$

Тогава $\exists$ път в L $q_0 \xrightarrow{a_1} \ldots \xrightarrow{a_n} f \in F$

$\Rightarrow \exists$ извод $q_0 \to a_1 q_1 \to \ldots \to a^n f b^m \to a^n b^m$, където за

$a_i = a$ използвайки правило $q_{i-1} \to a q_i$, иначе

$q_{i-1} \to q_i b$ и $|a^n b^m|_b$ = броя преходи $q_{i-1} \xrightarrow{b} q_i$

$\Rightarrow m = |\beta|_b$ и аналог. $n = |\beta|_a$ ✓

$\Rightarrow$ $L_{sort} = L(G) \Rightarrow L_{sort}$ е КСЕ

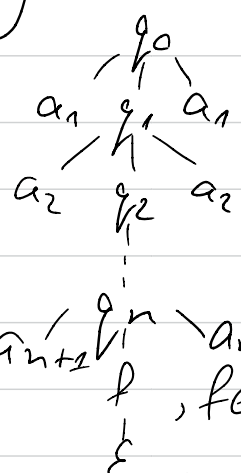
② Нека L-рег. ДСД, тк:

$L_{rev} = \{\alpha\alpha^{rev} \mid \alpha \in L\}$ е КСЕ

$G = (\Sigma, Q, q_0, P)$, $P = \{f \to \varepsilon \mid f \in F\} \cup \{q \to aqa \mid \delta(q,a) = \beta, a \in \Sigma\}$

$L(G) \stackrel{?}{=} L_{rev}$

(⊆) Нека $\alpha \in L(G)$. Тогава за α има-ме дърво на извод от вида



Освен това $\alpha = a_1 \dots a_n a_n \dots a_1$

и $\delta(q_i, a_{i+1}) = q_{i+1}$, $\delta(q_n, a_{n+1}) = f$, $f \in F$

$\Rightarrow$ в $A\nabla$ има $q_0 \xrightarrow{a_1} q_1 \to \dots \xrightarrow{a_n} q_n \xrightarrow{a_{n+1}} f \in F$

$\Rightarrow a_1 \dots a_{n+1} \in L \wedge \alpha = a_1 \dots a_{n+1} (a_1 \dots a_{n+1})^{rev}$

$\Rightarrow \alpha \in L_{rev}$

(2) $\alpha \in L_{rev} \Rightarrow \exists \beta \in L : L_{rev} = \beta\beta^{rev}$ и

$\exists$ нет $q_0 \xrightarrow{a_1} q_1 \to \dots \xrightarrow{a_n} q_n \in F$

$\Rightarrow \exists$ цикл $q_0 \to^{a_1} q_2^{a_1} \to \dots \to a_1 \dots a_n q_n^{a_n} \dots a_n \Rightarrow$

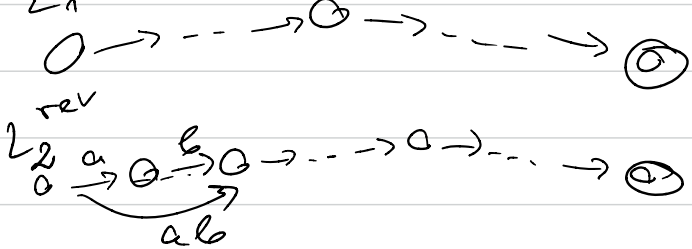
(3) $L = \{\alpha_1\alpha_2 \mid \alpha_1 \in L_1, \alpha_2 \in L_2, |\alpha_1| = 2|\alpha_2| \wedge$
L_1, L_2 – рег$\}$

$\varnothing \subset \mathcal{Q}, \varepsilon \in$ L – КСЕ

A_1 – авт. для L_1

A^2 – $A_1 \otimes A_2$

$L(A^2) = L_2^{rev}$

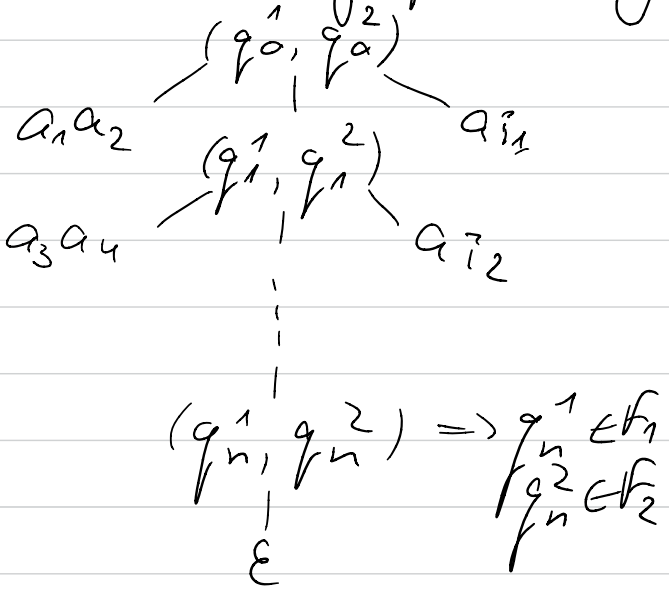


$G = (\Sigma, Q_1 \times Q_2, (q_0^1, q_0^2), P)$

$P = \{(f_1, f_2) \to \varepsilon \mid f_1 \in F_1, f_2 \in F_2\}$
$\cup \{(q_1, q_2) \to a_1 a_2 (p_1, p_2) a_3 \mid \delta_1^*(q_1, a_1 a_2) = p_1 \wedge \delta^2(q_2, a_3^{-1}) = \beta$

$L(G) = L$?

($\subseteq$) $\alpha \in L(G) \Rightarrow$ има горе на извода:



$\Rightarrow \alpha = a_1 \dots a_{2n} a_{i_n} \dots a_{i_1}$

$\delta_1^*(\delta_1^*(\dots(q_0^1, \underbrace{a_1 a_2}_{\beta}), a_3 a_4)\dots)a_{2n-1} a_{2n}) = f \in F_1$

$\Rightarrow a_1 \dots a_{2n} \in L_1 \wedge |\beta| = 2n$

$\delta^2(\delta^2(\dots(q_0^2, a_{i_1}) a_{i_2})\dots) a_{i_n}) = f \in F_2$

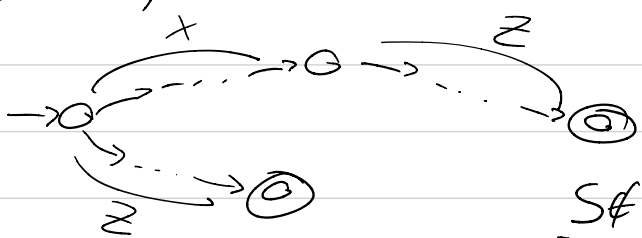
$\overset{rev}{\Rightarrow} a_{i_1} \dots a_{i_n} \in L_2 \Rightarrow a_{i_n} a_{i_{n-1}} \dots a_{i_1} \in L_2 \wedge \underbrace{|\beta_2| = n}_{\beta_2}$

$\Rightarrow a_1 \dots a_{2n} a_{i_n} \dots a_{i_1} = \beta_1 \beta_2$ $\quad |\beta_1| = 2n = 2|\beta_2|$

($\supseteq$) Аналог. $\Rightarrow \alpha \in L$

④ $L' = \{ zxz^{rev} \mid z \in L \wedge xz \in L \}$, L - рег.

Доказ, че L' - КСЕ



$S \notin Q$

$G = (\Sigma, Q \times Q \cup \{S\}, S, P)$

$P = \{ S \to (q_0, f) \mid f \in F \}$

$\cup \{ (q_1, q_2) \to a_1 (p_1, p_2) a_1 \mid a_1 \in \Sigma,\ \delta(q_1, a_1) = p_1 \wedge \delta(p_2, a_1) = q_2 \}$

$\cup \{ (f, q_1) \to q_0^1 \} \cup P_1$

$Q \times Q \times \{0, 1\}$ по-добре

$\{ (f, q_1, 0) \to (q_0, f_1, 1) \} \cup \{ (p_1, p_1, 1) \to \varepsilon \} \cup$

$\cup \{ (q_1, p_1, 1) \to a (p_2, p_1, 1) \mid \delta(q_1, a) = p_2 \}$

⑤ $I(L_1, L_2) = \{xyz \mid x z \in L_2, y \in L_1\}$

Решена при бобката